Practical No 3

Program: Write a program for sorting algorithms using appropriate knowledge representation and Reasoning techniques

```
# ----- Sorting Algorithms -----


def bubble_sort(arr):
    arr = arr.copy()   # avoid modifying original list
    n = len(arr)
    for i in range(n):
        for j in range(0, n - i - 1):
            if arr[j] > arr[j + 1]:
                arr[j], arr[j + 1] = arr[j + 1], arr[j]
    return arr



def selection_sort(arr):
    arr = arr.copy()
    n = len(arr)
    for i in range(n):
        min_index = i
        for j in range(i + 1, n):
            if arr[j] < arr[min_index]:
                min_index = j
        arr[i], arr[min_index] = arr[min_index], arr[i]
    return arr
```

```
def insertion_sort(arr):
    arr = arr.copy()
    for i in range(1, len(arr)):
        key = arr[i]
        j = i - 1
        while j >= 0 and key < arr[j]:
            arr[j + 1] = arr[j]
            j -= 1
        arr[j + 1] = key
    return arr


# ----- Knowledge Representation (Rules) -----

rules = {
    "small": insertion_sort,
    "medium": selection_sort,
    "large": bubble_sort
}


# ----- Reasoning Function -----

def choose_sort(arr):
    n = len(arr)
```

```
    if n <= 5:
        rule = "small"
    elif n <= 10:
        rule = "medium"
    else:
        rule = "large"

    print("Reasoning: Using", rule, "data rule")

    sorting_function = rules[rule]
    return sorting_function(arr)


# ----- Main Program -----

data = list(map(int, input("Enter numbers separated by space: ").split()))

sorted_data = choose_sort(data)

print("Sorted List:", sorted_data)
```

Home | Untitled20 | Untitled21

localhost:8888/notebooks/Untitled21.ipynb

jupyter Untitled21 Last Checkpoint: 11 minutes ago

File Edit View Run Kernel Settings Help

Trusted

Code

JupyterLab Python 3 (ipykernel)

```
    else:
        rule = "large"

    print("Reasoning: Using", rule, "data rule")

    sorting_function = rules[rule]
    return sorting_function(arr)


# ----- Main Program -----

data = list(map(int, input("Enter numbers separated by space: ").split()))

sorted_data = choose_sort(data)

print("Sorted List:", sorted_data)
```

```
Enter numbers separated by space:  2 5 88 34 99 4 14
Reasoning: Using medium data rule
Sorted List: [2, 4, 5, 14, 34, 88, 99]
```

[ ]:

Type here to search

24°C Sunny ENG 05:03 20-03-2026